

CAP 182
SS2
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PART A

PUBLICATION 1

Title	ORIC: Feature Interaction Detection through Online Random Interaction Chains for Click-Through Rate Prediction
Harvard Reference	Kou, Y., Lin, Q., & Gao, C. (2025). ORIC: Feature Interaction Detection through Online Random Interaction Chains for Click-Through Rate Prediction. ACM Transactions on Knowledge Discovery from Data, 19(3), Article 76.
Link	https://dl.acm.org/doi/10.1145/3717070
Modelling	Online Random Interaction Chains (ORIC) is a method used to find useful relationships between different categories of data to predict Click-Through Rate (CTR). It works in an online setting and uses random walks to identify these interactions.
Preprocessing Techniques	Categorical features (such as gender, product type, or location) were converted into numbers so the model could use them. Continuous features (such as age or price) were divided into groups or ranges. The data was split according to time, with older data used first and newer data used later. This was done to make the experiment more like real-time/online learning.
Data sets used	Avazu dataset (three datasets are available)

Key take aways	Choosing the right feature combinations can improve CTR prediction. The Avazu dataset has clear features that make it easier to understand how different features interact. For online learning, the data should be split based on time so that the model learns from older data before making predictions on newer data.
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PUBLICATION 2

Title	Exploiting Spiking Neural Networks for Click-Through Rate Prediction in Personalized Online Advertising Systems
Harvard Reference	Uruqi, A., & Viktoratos, I. (2025). Exploiting Spiking Neural Networks for Click-Through Rate Prediction in Personalized Online Advertising Systems. <i>Forecasting</i> , 7(3), Article 38.
Link	https://www.mdpi.com/2571-9394/7/3/38
Modelling	Spiking Neural Networks (SNNs) are AI models inspired by the human brain. They were compared with DCN-V2 and FinalMLP. SNNs use information over time and process information when an event happens.
Preprocessing Techniques	Categorical features were converted into numbers, numerical features were put on the same scale, and the data was divided into 70% training, 15% validation, and 15% testing.
Data sets used	Avazu and Digix datasets.

Key take aways	SNNs can achieve good CTR prediction accuracy while using less computing power. The Avazu dataset can be used to test new neural network models. AUC-ROC and accuracy are commonly used to measure how well CTR models perform.
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PUBLICATION 3 (PUBLICLY AVAILABLE DATASET)

Title	Avazu CTR Prediction Dataset
Harvard Reference	Avazu Inc. (2014). Avazu Click-Through Rate Prediction Challenge Dataset. Kaggle. Available at: https://www.kaggle.com/c/avazu-ctr-prediction/data
Link	https://www.kaggle.com/c/avazu-ctr-prediction/data
Number of instances	40,428,967 records (single training split; test split is also provided separately)
Number of feature columns	24 fields total (including the target label). After removing the ID and target, there are 22 feature fields (all categorical).
Number of classes (if applicable)	2 classes (click = 1, no click = 0).
Target Range (if applicable)	N/A
Origin	The data was collected by Avazu, a mobile advertising company. It contains 10 days of ad-click data from mobile devices, with 15 known features and 9 anonymous features. The data was anonymized to protect users' privacy.